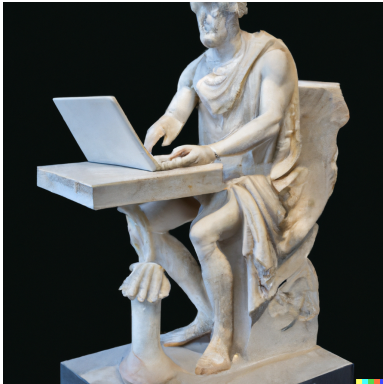


But first...

Trends and challenges in mechanics, according to machine learning:

An ancient Greek running a computer model, marble, ca. 200 BC



A futuristic Universal Testing Machine



Precalibrated model

Offline

Pretrained ML

Minimum data

A lot of data

Physics

Data

No data

Scarce data

Expensive model

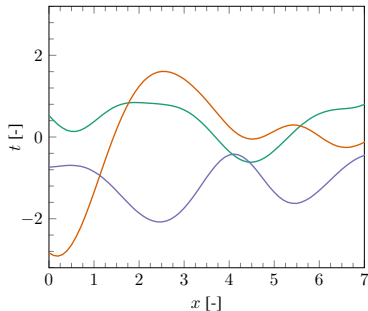
Online

Active learning

Active learning

Reducing sampling effort by only getting data when necessary:

- Expensive model response is unknown for most inputs $\Rightarrow$ **epistemic uncertainty**
- Bayesian machine learning can elegantly treat this problem



$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t} | \mathbf{0}, \mathbf{K})$$

$$K_{pq} = k(\mathbf{x}_p, \mathbf{x}_q) = \sigma_f^2 \exp\left(-\frac{1}{2\ell^2} \|\mathbf{x}_p - \mathbf{x}_q\|^2\right)$$

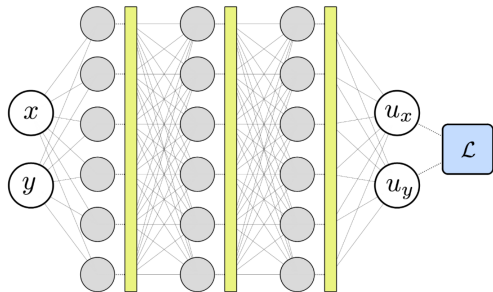
variance scaling

length scale

Physics-informed neural networks

Neural networks as PDE solvers:

- BC/IC values treated as conventional observations
- Residual of strong-form PDE at collocation points added to loss function



[Haghighat et al (2020), arXiv:2003.02751v2]

$$\sigma_{ij,j} + f_i = 0$$

$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij}$$

$$\varepsilon_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i})$$

$$\mathcal{L} = |\sigma_{ij,j} + f_i|_{\Omega} + |u - u^*|_{\Gamma_u}$$

Col. point
PDE loss

Dirichlet
loss

Offline

Physics

Data

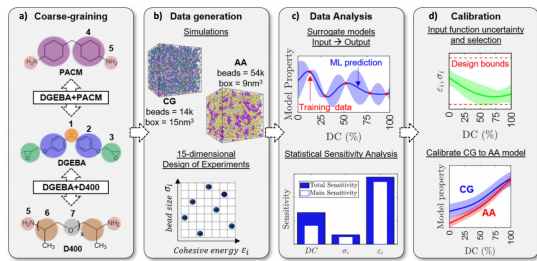
Online

Optimization and discovery with ML

Learning architectures for material design:

- Design exploration, Bayesian optimization
- Generative machine learning

Bayesian optimization for polydispersity in branched polymers



[Andrea Giuntoli's group (RUG)]

